



NUBTK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

KNOWLEDGE FOR AN ENLIGHTENING FUTURE

[Practice Problems]

Semester	:	Fall 2025
Course Title	:	Object - Oriented Programming II
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Sections	:	3A, 3B, 3C, 3D

Here I have provided 50 practice problems. These are only sample exercises to help you improve your Java programming skills. If you practice these properly, you will do well in Java.

♦ Important Note:

These problems are not suggestions for the exam, and you will not get any common questions from this list.

They are purely for practice, to build your confidence and understanding.

I strongly recommend all of you to practice each problem sincerely.

If you put in the effort, your programming knowledge will definitely improve.

Best of luck!

Start practicing.

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Problem Set 1

Problem 1: Hello World

Question: Write a Java program that prints "Hello, World!" to the console.

Sample Input: None

Sample Output: Hello, World!

Problem 2: Arithmetic Operations

Question: Write a program that takes two integers, performs addition, subtraction, multiplication, and division, and displays the results.

Sample Input: a = 10, b = 5

Sample Output:

Addition: 15

Subtraction: 5

Multiplication: 50

Division: 2

Problem 3: Area of a Circle

Question: Write a Java program to calculate the area of a circle with a given radius.

Sample Input: radius = 7.5

Sample Output: Area of Circle: 176.71

Problem 4: Temperature Conversion

Question: Convert temperature from Celsius to Fahrenheit using the formula $F = (C * 9/5) + 32$.

Sample Input: Celsius = 25

Sample Output: Temperature in Fahrenheit: 77.0

Problem 5: Swap Two Numbers

Question: Swap two numbers using a temporary variable.

Sample Input: x = 10, y = 20

Sample Output: After swap, x = 20, y = 10

Problem 6: Simple Interest Calculator

Question: Calculate the simple interest based on principal, rate, and time.

Sample Input: principal = 10000, rate = 5, time = 3

Sample Output: Simple Interest: 1500.0

Problem 7: Determine Odd or Even

Question: Write a program to check if a number is odd or even.

Sample Input: num = 29

Sample Output: 29 is odd.

Problem 8: Find Maximum of Two Numbers

Question: Write a program to find the maximum of two numbers.

Sample Input: a = 15, b = 25

Sample Output: The maximum number is: 25

Problem 9: Compute Average of Three Numbers

Question: Write a Java program to compute the average of three numbers.

Sample Input: num1 = 8.5, num2 = 7.3, num3 = 9.8

Sample Output: Average: 8.533333333333333

Problem 10: Sum of Digits

Question: Write a program to calculate the sum of digits of a given number.

Sample Input: number = 1234

Sample Output: Sum of digits: 10

Problem Set 2

Problem 1:

Question: Write a Java program to find the sum of all elements in an integer array.

Sample Input:

Enter number of elements: 5

Enter elements: 1 2 3 4 5

Sample Output: Sum of array elements: 15

Problem 2:

Question: Write a Java program to find the largest element in an integer array.

Sample Input:

Enter number of elements: 4

Enter elements: 3 5 9 2

Sample Output: Largest element: 9

Problem 3:

Question: Write a Java program to count the occurrences of a specific element in an array.

Sample Input:

Enter number of elements: 6

Enter elements: 1 2 2 3 2 4

Enter element to count: 2

Sample Output: Occurrences of 2: 3

Problem 4:

Question: Write a Java program to reverse the elements in an array.

Sample Input:

Enter number of elements: 5

Enter elements: 10 20 30 40 50

Sample Output: Reversed array: 50 40 30 20 10

Problem 5:

Question: Write a Java program to check if an array is sorted in ascending order.

Sample Input:

Enter number of elements: 5

Enter elements: 1 2 3 4 5

Sample Output: Array is sorted

Problem 6:

Question: Write a Java program to merge two integer arrays into a single array.

Sample Input:

Enter number of elements in first array: 3

Enter elements: 1 3 5

Enter number of elements in second array: 3

Enter elements: 2 4 6

Sample Output: Merged array: 1 3 5 2 4 6

Problem 7:

Question: Write a Java program to remove duplicate elements from an integer array.

Sample Input:

Enter number of elements: 6

Enter elements: 1 2 2 3 4 4

Sample Output: Array without duplicates: 1 2 3 4

Problem 8:

Question: Write a Java program to find the second largest element in an array.

Sample Input:

Enter number of elements: 5

Enter elements: 7 5 8 9 6

Sample Output: Second largest element: 8

Problem 9:

Question: Write a Java program to rotate an array to the right by one position.

Sample Input:

Enter number of elements: 4

Enter elements: 10 20 30 40

Sample Output: Rotated array: 40 10 20 30

Problem 10:

Question: Write a Java program to find a pair of elements in an array that sum up to a specific number.

Sample Input:

Enter number of elements: 5

Enter elements: 1 2 3 4 5

Enter target sum: 7

Sample Output: Pair found: (2, 5)

Problem Set 3

Problem 1:

Question: Write a Java program to find the sum of all elements in an integer array.

Sample Input:

Enter number of elements: 5

Enter elements: 1 2 3 4 5

Sample Output: Sum of array elements: 15

Problem 2:

Question: Write a Java program to check if a given number is even or odd.

Sample Input:

Enter a number: 7

Sample Output: Odd

Problem 3:

Question: Write a Java program to find the largest of three numbers.

Sample Input:

Enter three numbers: 5 9 3

Sample Output: Largest number: 9

Problem 4:

Question: Write a Java program to check if a given year is a leap year.

Sample Input:

Enter a year: 2024

Sample Output: Leap Year

Problem 5:

Question: Write a Java program to calculate the factorial of a number.

Sample Input:

Enter a number: 5

Sample Output: Factorial: 120

Problem 6:

Question: Write a Java program to check if a number is prime.

Sample Input:

Enter a number: 11

Sample Output: Prime

Problem 7:

Question: Write a Java program to find the sum of the digits of a number.

Sample Input:

Enter a number: 1234

Sample Output: Sum of digits: 10

Problem 8:

Question: Write a Java program to print the multiplication table of a given number.

Sample Input:

Enter a number: 3

Sample Output:

3 x 1 = 3

3 x 2 = 6

3 x 3 = 9

...

3 x 10 = 30

Problem 9:

Question: Write a Java program to count the number of vowels in a given string.

Sample Input:

Enter a string: Hello World

Sample Output: Number of vowels: 3

Problem 10:

Question: Write a Java program to perform basic calculations (addition, subtraction, multiplication, division) based on user input.

Sample Input:

Enter first number: 8

Enter operator (+, -, *, /): *

Enter second number: 5

Sample Output: Result: 40

Problem Set 4

Each problem (From 1 to 10) includes a **Main Class** and an **Incomplete Class**. You must **complete the incomplete class** so the program runs correctly.

Problem 1

Main Class

```
public class StudentResult {  
    public static void main(String[] args) {  
        Student s1 = new Student("Rahim", 85, 90, 88);  
        double avg = s1.getAverage();  
        System.out.println("Average marks: " + avg);  
    }  
}
```

Incomplete Class

```
public class Student {  
    String name;  
    int m1, m2, m3;  
  
    // incomplete  
}
```

Problem 2

Main Class

```
public class RectangleTest {  
    public static void main(String[] args) {  
        Rectangle r = new Rectangle(5.0, 10.0);  
        System.out.println("Area: " + r.area());  
    }  
}
```

Incomplete Class

```
public class Rectangle {  
    double length;  
    double width;  
  
    // incomplete  
}
```

Problem 3

Main Class

```
public class EmployeeSalary {  
    public static void main(String[] args) {  
        Employee e = new Employee("Shovon", 30000, 5000);  
        System.out.println("Total Salary: " + e.totalSalary());  
    }  
}
```

Incomplete Class

```
public class Employee {  
    String name;  
    double basic;
```



```
double bonus;  
  
// incomplete  
}
```

Problem 4**Main Class**

```
public class BookTest {  
    public static void main(String[] args) {  
        Book b = new Book("Java Programming", 550);  
        b.displayInfo();  
    }  
}
```

Incomplete Class

```
public class Book {  
    String title;  
    double price;  
  
// incomplete  
}
```

Problem 5**Main Class**

```
public class CircleMain {  
    public static void main(String[] args) {  
        Circle c = new Circle(7.0);  
        System.out.println("Circumference: " + c.circumference());  
    }  
}
```

Incomplete Class

```
public class Circle {  
    double radius;  
  
// incomplete  
}
```

Problem 6**Main Class**

```
public class ProductTest {  
    public static void main(String[] args) {  
        Product p = new Product("Laptop", 50000, 5);  
        System.out.println("Total Cost: " + p.totalCost());  
    }  
}
```

Incomplete Class

```
public class Product {  
    String name;  
    double price;  
    int quantity;
```

```
// incomplete
}
```

Problem 7**Main Class**

```
public class CarMain {
    public static void main(String[] args) {
        Car c = new Car("Toyota", 2020);
        c.display();
    }
}
```

Incomplete Class

```
public class Car {
    String brand;
    int model;
```

```
// incomplete
}
```

Problem 8**Main Class**

```
public class BankMain {
    public static void main(String[] args) {
        BankAccount b = new BankAccount("A123", 15000);
        b.deposit(5000);
        System.out.println("Balance: " + b.getBalance());
    }
}
```

Incomplete Class

```
public class BankAccount {
    String accNo;
    double balance;
```

```
// incomplete
}
```

Problem 9**Main Class**

```
public class LaptopMain {
    public static void main(String[] args) {
        Laptop l = new Laptop("HP", "i5", 8);
        l.showSpecs();
    }
}
```

Incomplete Class

```
public class Laptop {
    String brand;
    String processor;
    int ram;
```

```
// incomplete  
}
```

Problem 10**Main Class**

```
public class MarksTest {  
    public static void main(String[] args) {  
        Marks m = new Marks(75, 80, 85);  
        System.out.println("Total: " + m.total());  
    }  
}
```

Incomplete Class

```
public class Marks {  
    int a, b, c;  
  
    // incomplete  
}
```

Problem Set 5 (Trace-and-Output Problems)

Here are **10 questions**, where you must **trace and find the output** of a Java program. Each question includes a **full code snippet** with small logic twists so you must carefully analyze the flow.

Problem 1

```
public class TraceTest1 {
    public static void main(String[] args) {
        int a = 0;
        int b = 5;

        if (a == 0) {
            b = b + 2;
        } else {
            b = b - 2;
        }

        int[] arr = new int[b];
        for (int i = 0; i < b; i++) {
            arr[i] = i * 2;
        }

        for (int x : arr) {
            System.out.print(x + " ");
        }
    }
}
```

Problem 2

```
public class TraceTest2 {
    public static void main(String[] args) {
        int x = 3, y = 0;

        y = (x % 2 == 1) ? x + 2 : x - 2;

        int size = x - 1;
        int[] arr = new int[size];

        for (int i = 0; i < size; i++) {
            arr[i] = y - i;
        }

        for (int v : arr) {
            System.out.print(v + " ");
        }
    }
}
```

Problem 3

```
public class TraceTest3 {
    public static void main(String[] args) {
        int n = 1;
        int k = (n == 1) ? 3 : 2;

        int[] arr = new int[k];
        for (int i = 0; i < k; i++) {
            arr[i] = n + i * 2;
        }

        for (int x : arr) {
            System.out.print(x + " ");
        }
    }
}
```

Problem 4

```
public class TraceTest4 {
    public static void main(String[] args) {
        int a = 2, b = 2;

        if (b == 2) a = a + b;
        else a = a - 1;

        int size = a / 2;
        int[] arr = new int[size];

        for (int i = 0; i < size; i++) {
            arr[i] = a - i;
        }

        for (int t : arr) {
            System.out.print(t + " ");
        }
    }
}
```

Problem 5

```
public class TraceTest5 {
    public static void main(String[] args) {
        int x = 0, y = 0;

        if (x == 0 && y == 0) x = 2;
        y = (x > 0) ? x + 1 : 1;

        int[] arr = new int[y];
        for (int i = 0; i < y; i++) {
            arr[i] = x + i;
        }
    }
}
```

```
    }

    for (int n : arr) {
        System.out.print(n + " ");
    }
}
}
```

Problem 6

```
public class TraceTest6 {
    public static void main(String[] args) {
        int a = 4;
        int b = a % 3;

        if (b == 1) b += 2;
        else b += 1;

        int[] arr = new int[b];

        for (int i = 0; i < b; i++) {
            arr[i] = b - i;
        }

        for (int x : arr) {
            System.out.print(x + " ");
        }
    }
}
```

Problem 7

```
public class TraceTest7 {
    public static void main(String[] args) {
        int n = 5;
        int m = (n > 3) ? 2 : 1;

        int size = n - m;
        int[] arr = new int[size];

        for (int i = 0; i < size; i++) {
            arr[i] = i + m;
        }

        for (int v : arr) {
            System.out.print(v + " ");
        }
    }
}
```

Problem 8

```
public class TraceTest8 {
```

```
public static void main(String[] args) {
    int p = 3, q = 1;

    if (p == 3) q = q + 3;
    else q = q + 1;

    int[] arr = new int[q];

    for (int i = 0; i < q; i++) {
        arr[i] = p * i;
    }

    for (int t : arr) {
        System.out.print(t + " ");
    }
}
```

Problem 9

```
public class TraceTest9 {
    public static void main(String[] args) {
        int a = 1;
        int b = (a == 1) ? 4 : 2;

        int size = b - 1;
        int[] arr = new int[size];

        for (int i = 0; i < size; i++) {
            arr[i] = a + b + i;
        }

        for (int n : arr) {
            System.out.print(n + " ");
        }
    }
}
```

Problem 10

```
public class TraceTest10 {
    public static void main(String[] args) {
        int x = 2, y = 5;

        if (y % 2 == 1) x++;
        else y--;

        int size = x;
        int[] arr = new int[size];

        for (int i = 0; i < size; i++) {
            arr[i] = y - i;
        }
    }
}
```

```
    }  
  
    for (int v : arr) {  
        System.out.print(v + " ");  
    }  
}  
}
```
